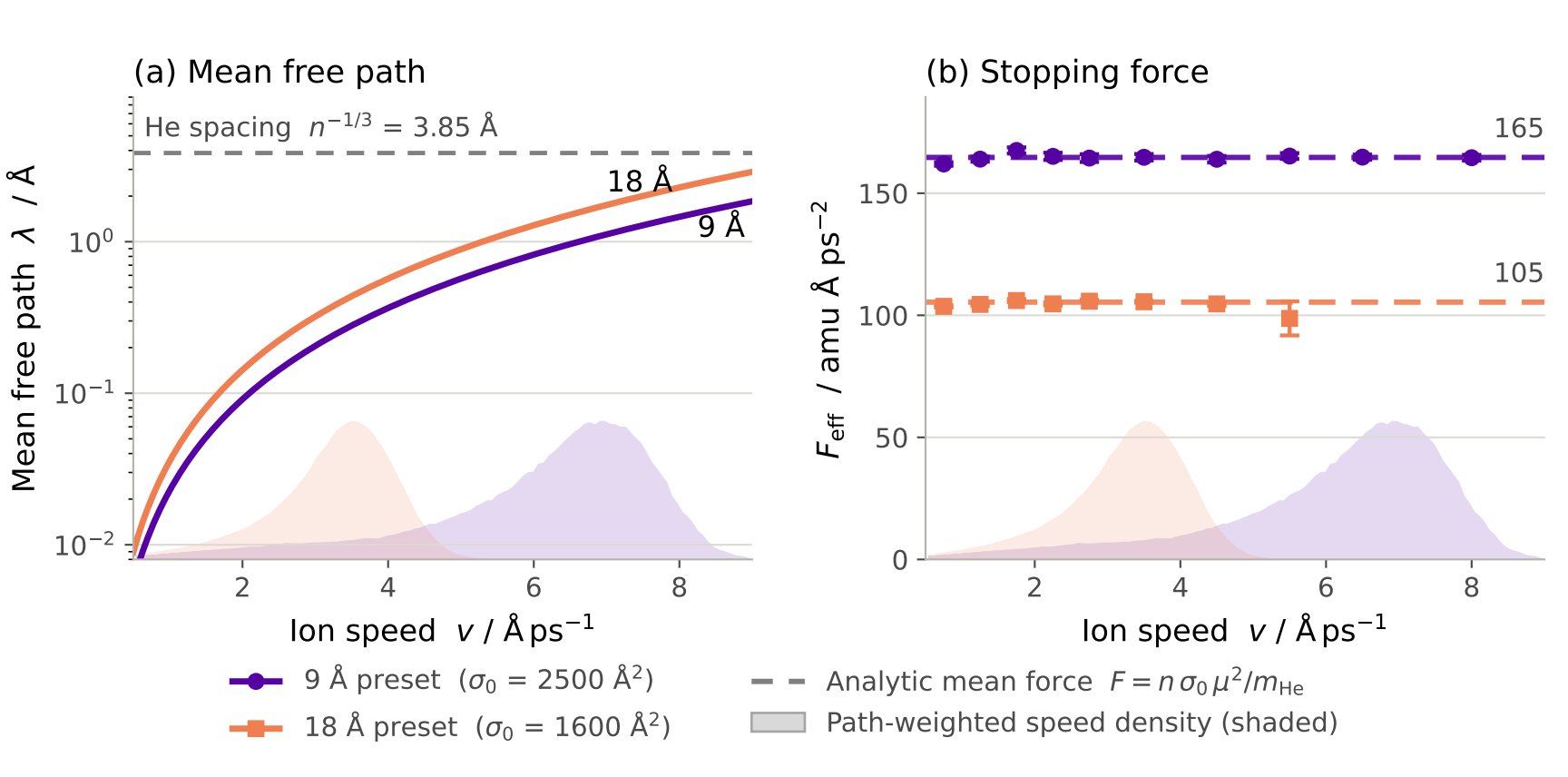
(a) Mean free path
He spacing $n^{-1/3}$ = 3.85 Å
18 Å
9 Å
Mean free path λ / Å
10^{0}
10^{-1}
10^{-2}
2
4
6
8
Ion speed v / Å ps^{-1}
(b) Stopping force
165
105
150
100
50
0
F_{eff} / amu Å ps^{-2}
2
4
6
8
Ion speed v / Å ps^{-1}
9 Å preset (σ_0 = 2500 Å^2)
18 Å preset (σ_0 = 1600 Å^2)
Analytic mean force $F = n\,\sigma_0\,\mu^2/m_{\mathrm{He}}$
Path-weighted speed density (shaded)